Stockholm Biochar Project

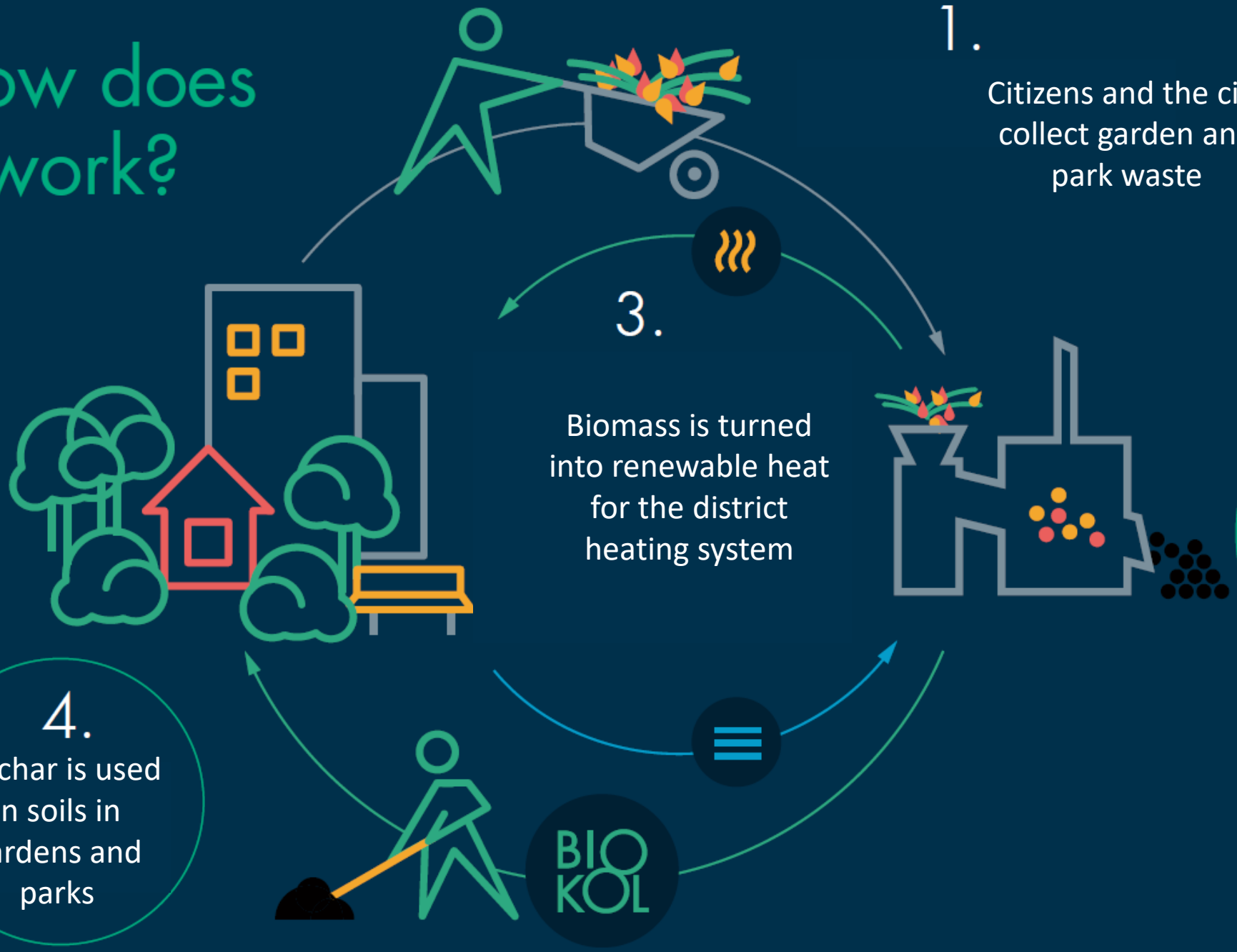
Mattias Gustafsson
Project manager for Stockholm Biochar Project
Founder of the Swedish Biochar Association
Vice CEO of EcoTopic AB



Winning Mayors Challenge

- Innovation competition for cities
- Bloomberg Philanthropies founder and financer
- 155 participating cities in Europe in 2014, 5 winners
- Project support and collaboration between the winning cities for 3 years

How does it work?



1.

Citizens and the city collect garden and park waste

2.

Biomass is turned into biochar through public and private collaboration

3.

Biomass is turned into renewable heat for the district heating system

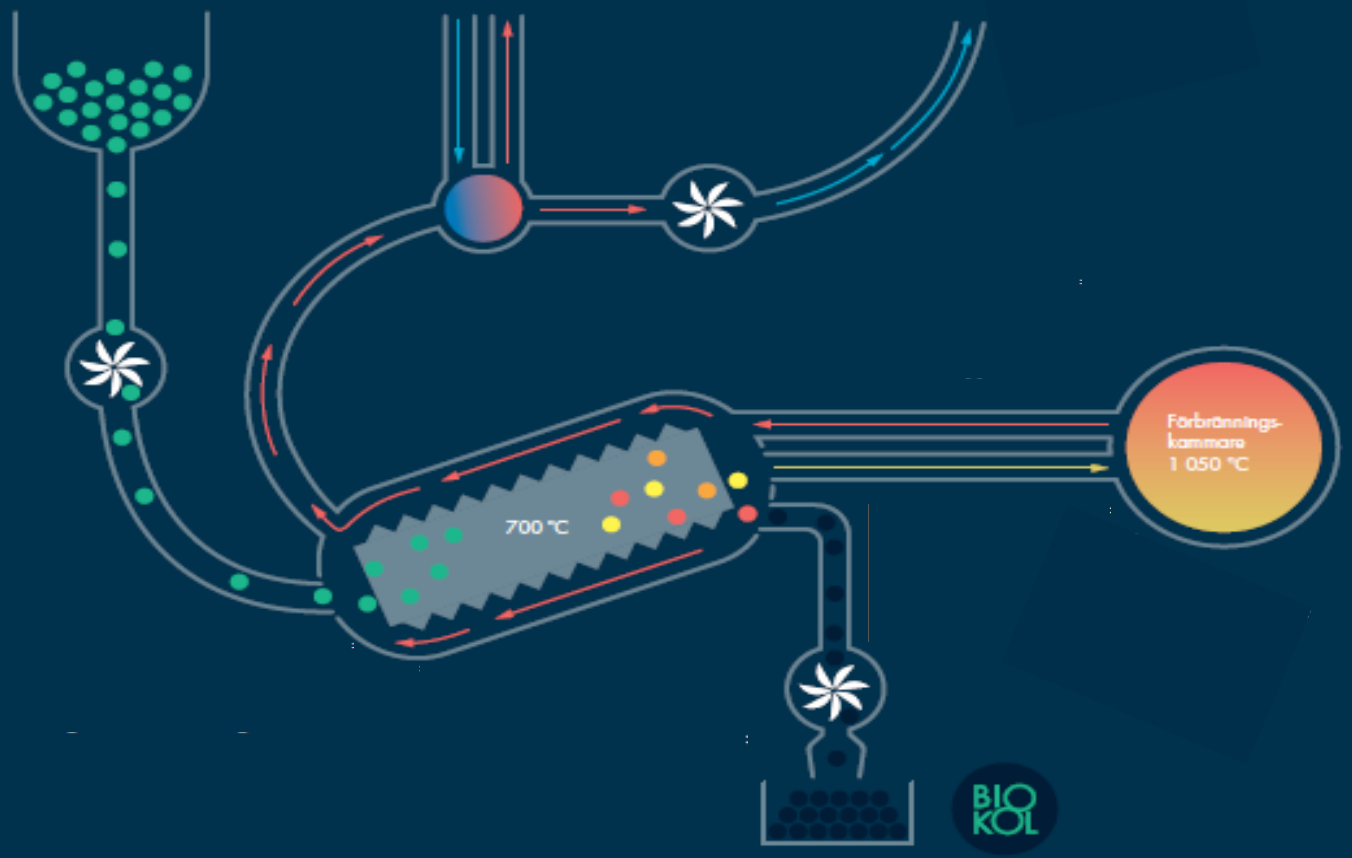
4.

Biochar is used in soils in gardens and parks

Technology

- Pyrolysis
  - Oxygen free environment
  - Carbonisation
- Heat
  - Pyrolysis gas
  - Low emissions
- Biochar
  - Char
  - Ash

The Stockholmer's garden waste becomes a part of the cycle





Pilot plant

Renewable heat to 80 apartments

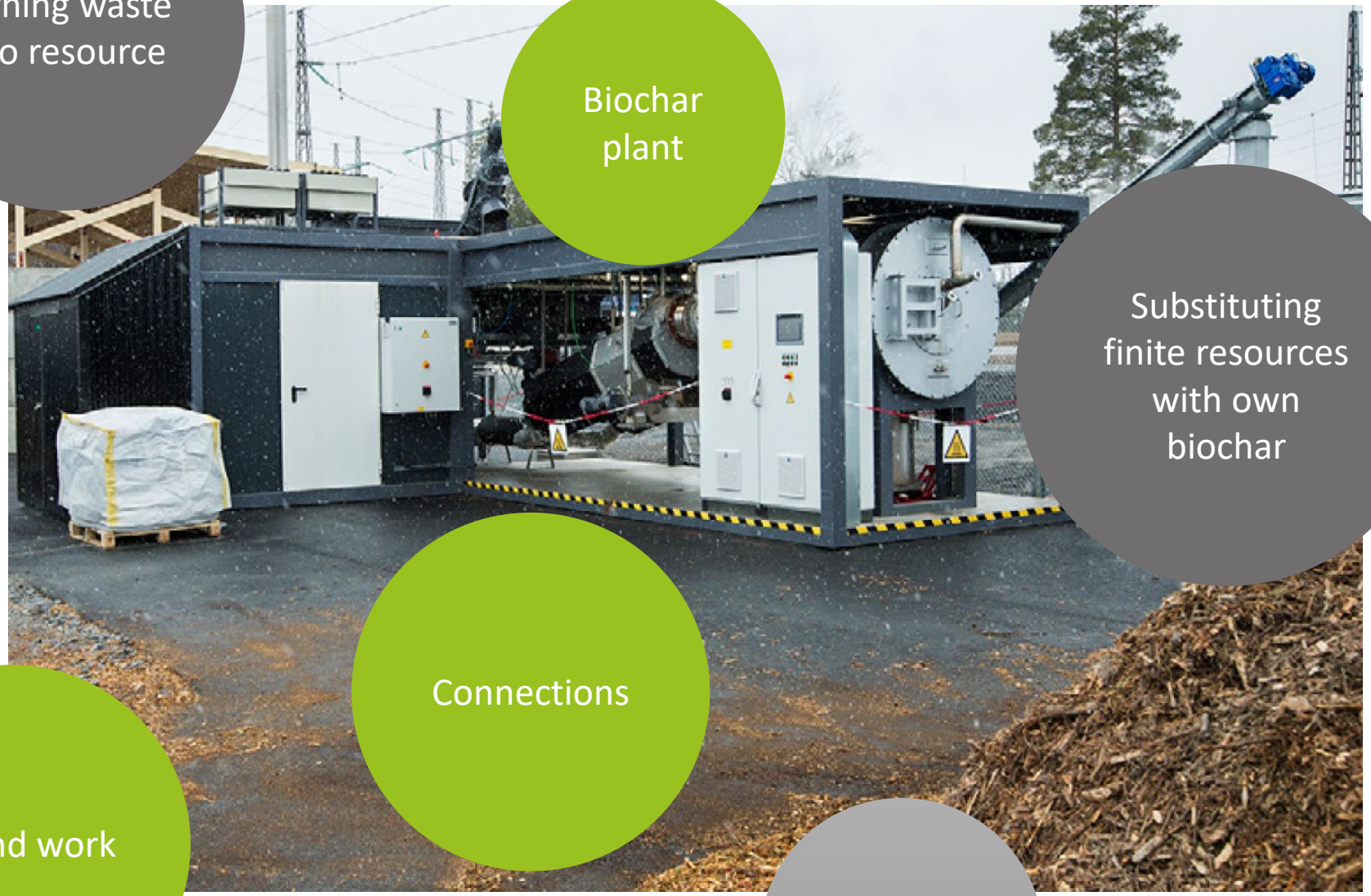
Biochar corresponding to CO_2-emissions from 700 cars

Full scale

Renewable heat to 400 apartments

Biochar corresponding to CO_2-emissions from 3500 cars

Can you make a profit on biochar?



Turning waste into resource

Biochar plant

Substituting finite resources with own biochar

Connections

Ground work

Selling heat

TOPIC
DO GOOD BE GOOD FEEL GOOD

Making biochar known

- ~1500 study visitors
- ~300 000 Stockholmers receiving direct information of the project
- 300 new urban plant beds in Stockholm with biochar in 2018
- 1500 Stockholmers have collected biochar
- Award for best recycling facility in Sweden in 2018



Stockholm Biochar – the Experiment



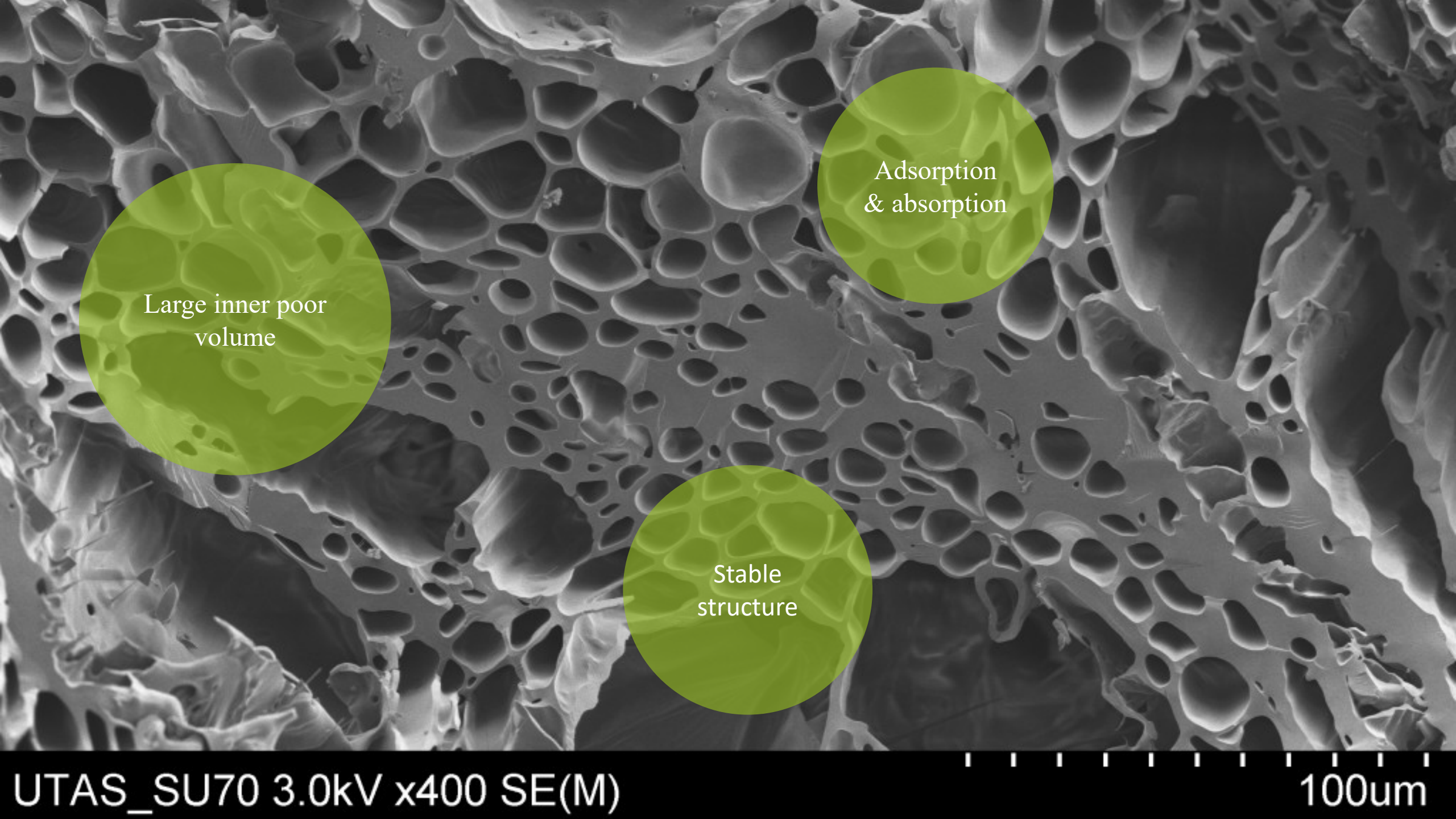
- Inspiration from the Big Biochar Experiment in Oxford started by Cécile Girardin (Oxford Biochar)
- Participants: 65 (2016), 125 (2017)
- Varying results but increasing knowledge of biochar

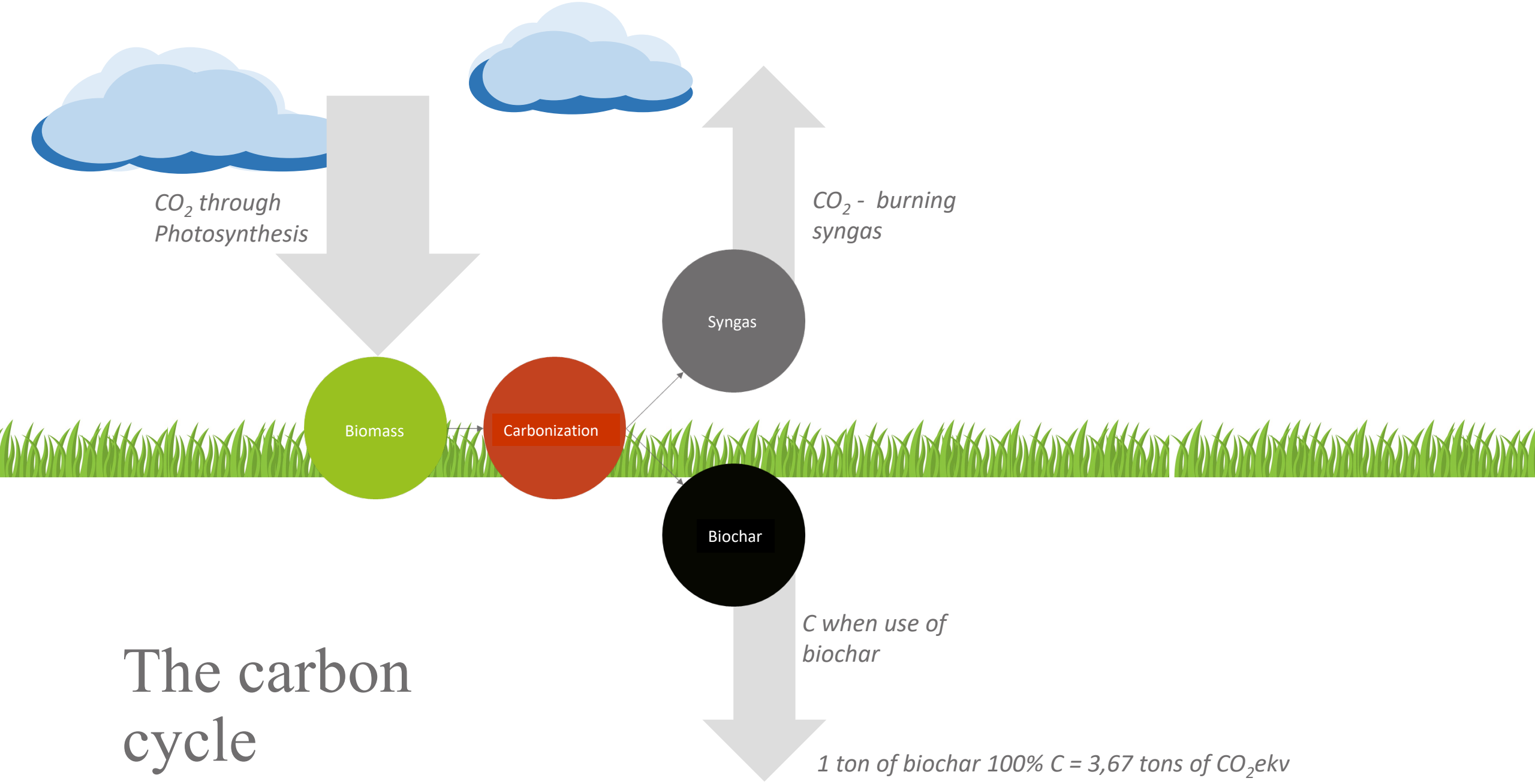









Large inner poor volume
Adsorption & absorption
Stable structure
UTAS_SU70 3.0kV x400 SE(M)
100um



The carbon cycle



Learnings

- Choose site based on costs
- There are only a few biochar plant suppliers
- Pre-treatment of garden waste is necessary
- Need of education

Continuing in 2019 and 2020



Making the pilot profitable

Stockholmers can keep collecting biochar

Evaluate

Deciding how to move forward

TOPIC
DO GOOD BE GOOD FEEL GOOD

The urban carbon sink

Urban tree beds with structural soil and biochar


Region with 3,5 million citizens in the year of 2030


100 yrs


15 yrs


50 yrs


30 yrs


6 yrs


100 yrs

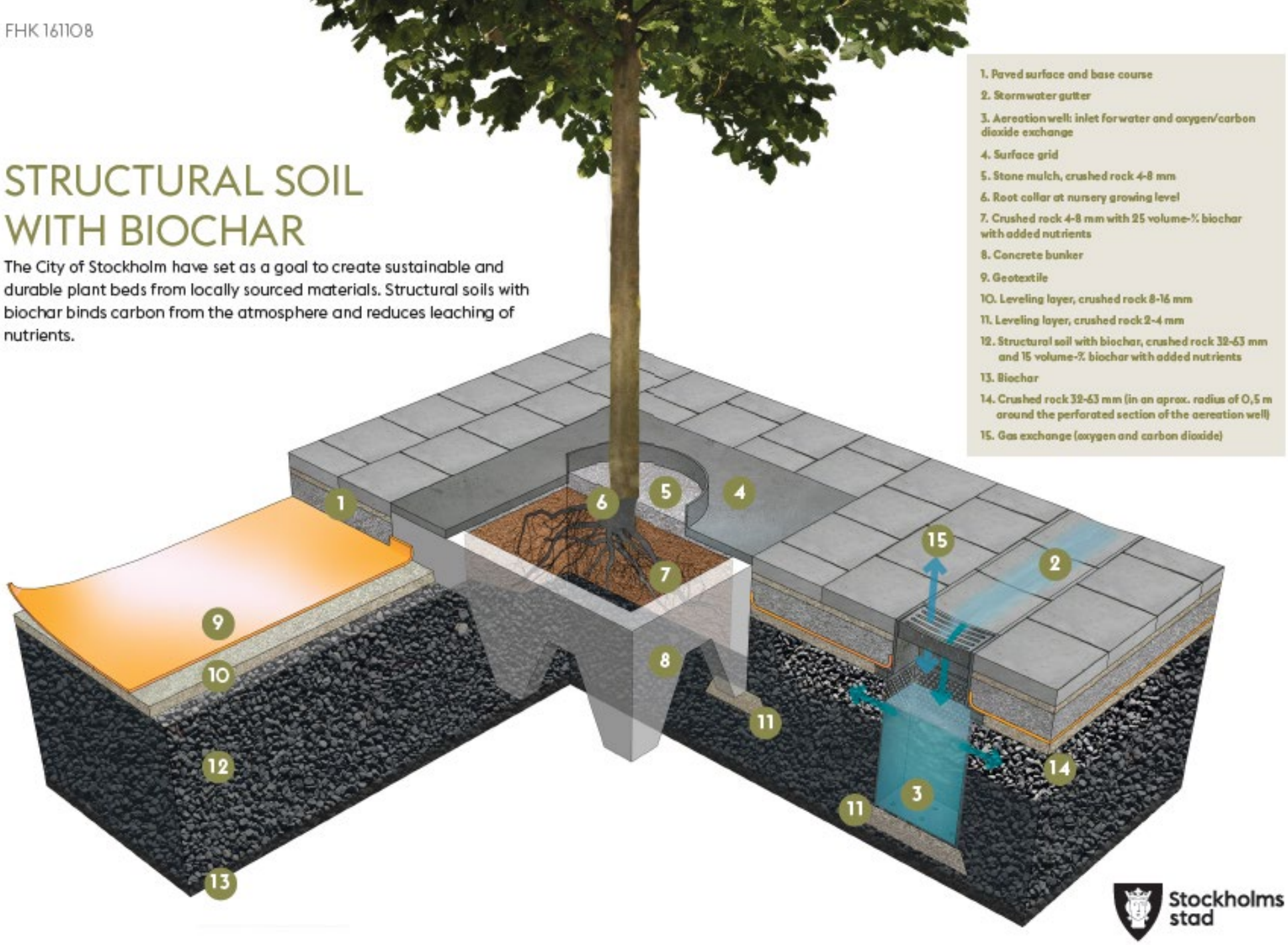

15 yrs


75 yrs

FHK 161108

STRUCTURAL SOIL WITH BIOCHAR

The City of Stockholm have set as a goal to create sustainable and durable plant beds from locally sourced materials. Structural soils with biochar binds carbon from the atmosphere and reduces leaching of nutrients.



1. Paved surface and base course
2. Stormwater gutter
3. Aeration well: inlet for water and oxygen/carbon dioxide exchange
4. Surface grid
5. Stone mulch, crushed rock 4-8 mm
6. Root collar at nursery growing level
7. Crushed rock 4-8 mm with 25 volume-% biochar with added nutrients
8. Concrete bunker
9. Geotextile
10. Leveling layer, crushed rock 8-16 mm
11. Leveling layer, crushed rock 2-4 mm
12. Structural soil with biochar, crushed rock 32-63 mm and 15 volume-% biochar with added nutrients
13. Biochar
14. Crushed rock 32-63 mm (in an aprox. radius of 0,5 m around the perforated section of the aeration well)
15. Gas exchange (oxygen and carbon dioxide)

Stockholms stad

TOPIC
DO GOOD BE GOOD FEEL GOOD









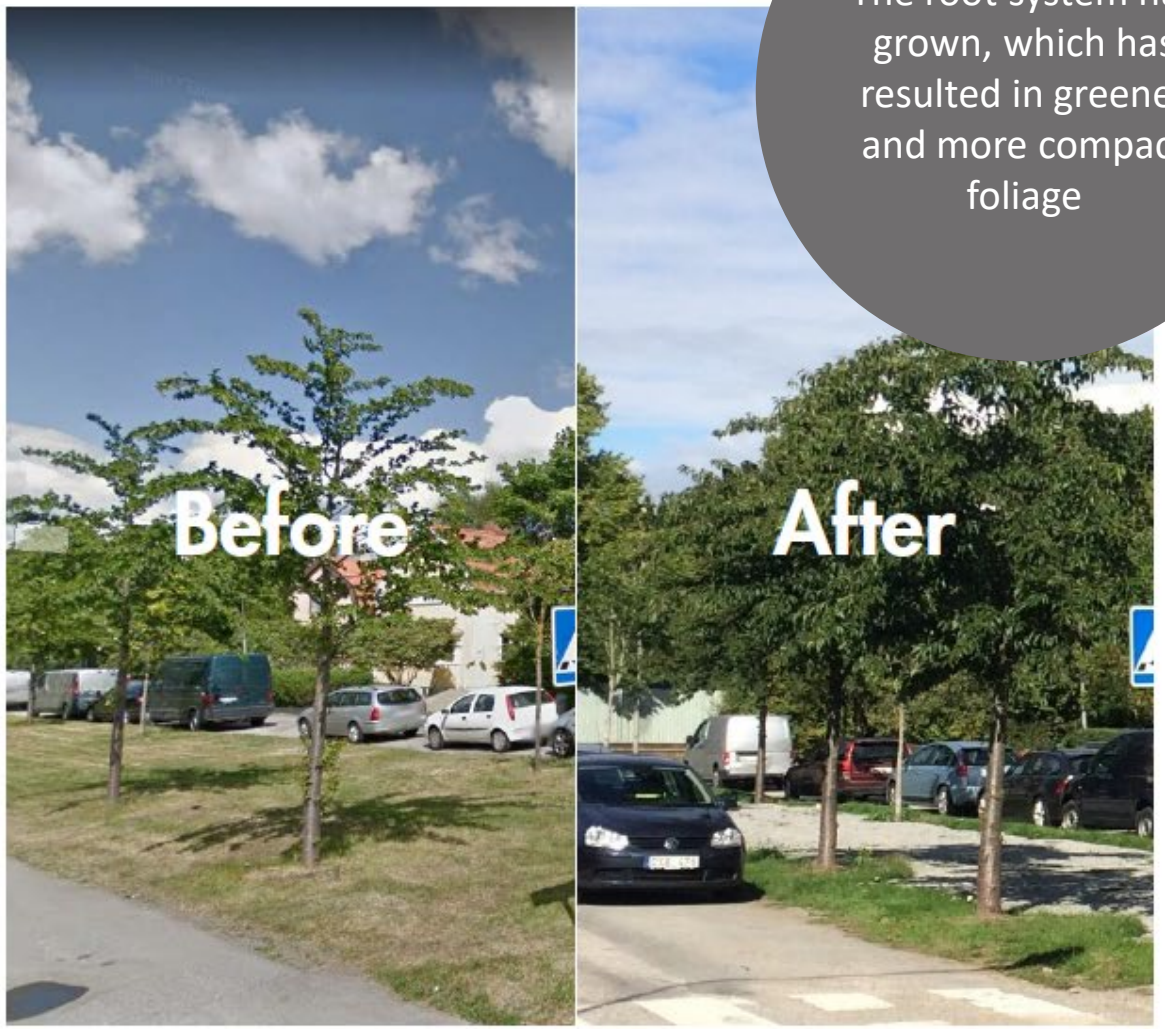
Nybrogatan 2015,
biochar & macadam


Macadam 32-63 mm
with 15% nutrient
enriched biochar


Nybrogatan
2016, magnolia

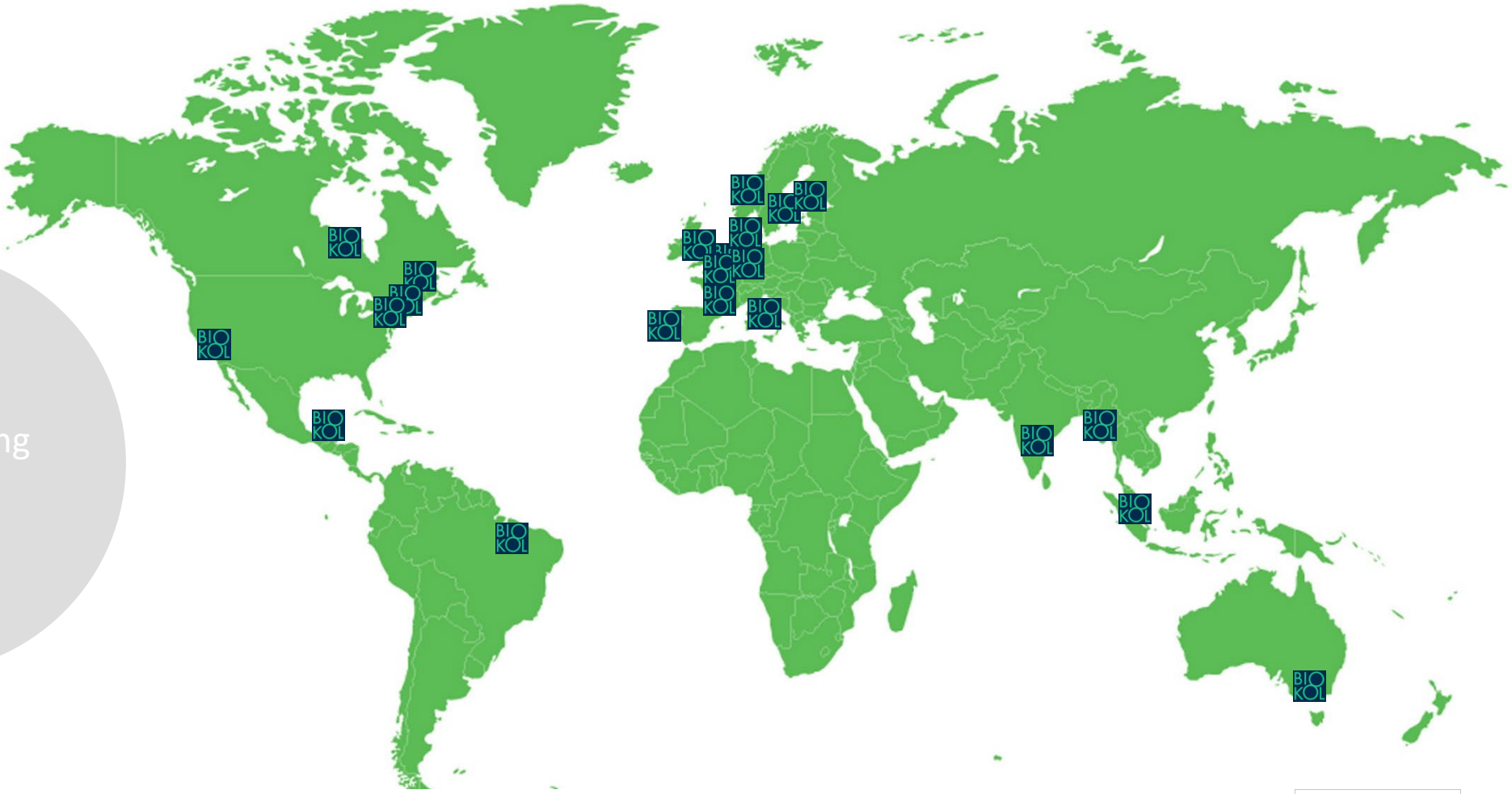
After one year with biochar





The root system has grown, which has resulted in greener and more compact foliage

Potential global replication



Access to biomass

Understanding biochar

Need for energy

Biochar production in Sweden



Stockholm Biochar Project

Hjelmsäters Egendom

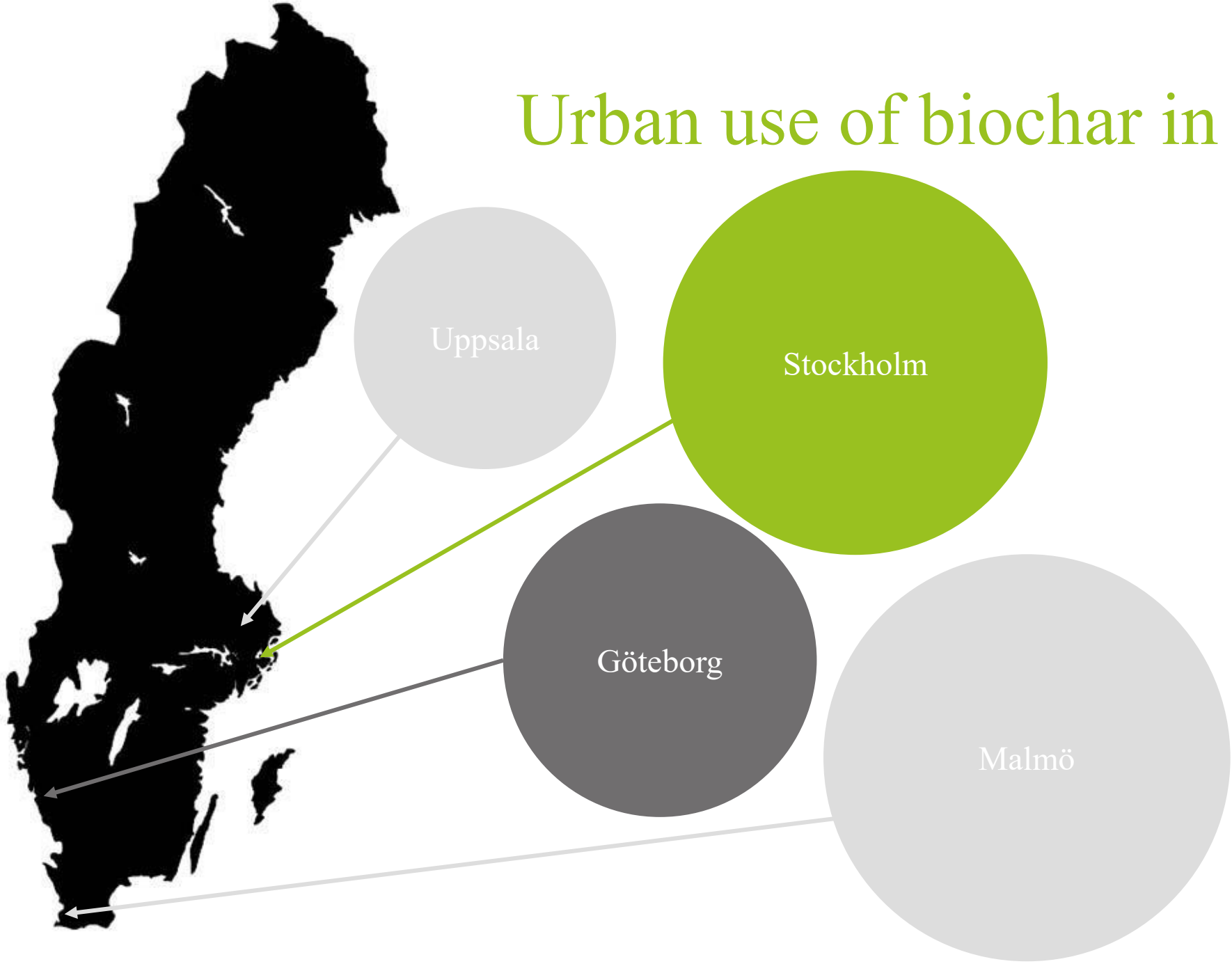
Lindeborgs gård

ETC

VegTech

Skånefrö

Urban use of biochar in Sweden



Needs for the Swedish market





Nordic Biochar Conference 2019

- 16th – 17th of October 2019
- KTH – Stockholm
- 10th of September
- http://www.nordicbiochar.org/

Thank you for your attention!

Mattias Gustafsson

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